

Вариант 3;

Линейный сплайн, $m = 1$

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Clear[P, PP, PPP, A, B, eq, koef, T, X, F, NN, n, xx, spl1, spl2, spl3];
1. Задаем начальные данные;
Subscript[x, 0] = 1; Subscript[x, 1] = 3; Subscript[x, 2] = 5; Subscript[x, 3] = 8;
Subscript[f, 0] = 4; Subscript[f, 1] = -20; Subscript[f, 2] = 45; Subscript[f, 3] = 76;
n = 3;

2. Формируем линейный интерполяционный сплайн;
Subscript[P, k_][t_] = Subscript[f, k] + (t - Subscript[x, k])*Subscript[A, k];
eq1 = Table[Subscript[P, k][Subscript[x, k + 1]] == Subscript[P, k + 1][Subscript[x, k + 1]], {k, 0, n - 1}];
eq2 = {Subscript[P, n][Subscript[x, n]] == Subscript[f, n]};
eq = Join[eq1, eq2];
koef = Solve[eq, {}];
spl1 = Table[Subscript[P, k][t] /. koef, {k, 0, n - 1}] // Expand // Flatten
Table[(spl1[[k]] /. t -> Subscript[x, k - 1]) == Subscript[f, k - 1], {k, 1, n}]
(spl1[[n]] /. t -> Subscript[x, n]) == Subscript[f, n]
{4 + 2 A0 == -20, -20 + 2 A1 == 45, 45 + 3 A2 == 76}

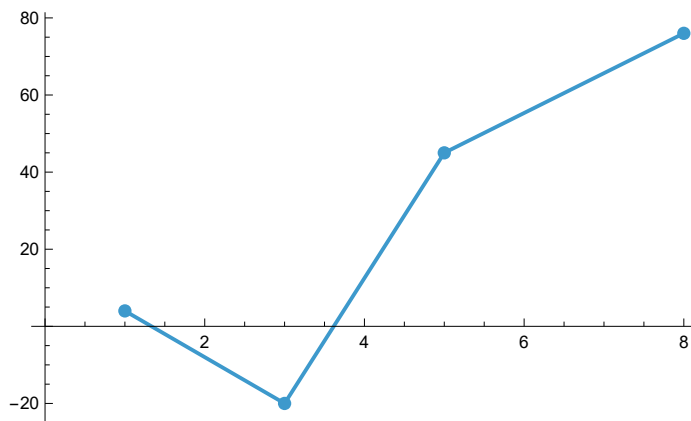
{{A0 -> -12, A1 -> 65/2, A2 -> 31/3}}

{16 - 12 t, -235/2 + 65 t/2, -20/3 + 31 t/3}

{True, True, True}

True
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GrA = ListPlot[Table[{Subscript[x, i], Subscript[f, i]}, {i, 0, n}], PlotStyle -> {PointSize[Large]};
GrB = Table[Plot[spl1[[k]], {t, Subscript[x, k - 1], Subscript[x, k]}, {k, 1, n}];
Show[GrA, GrB]
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Квадратичный сплайн, $m = 2$, $S'(a) = 0$

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Clear[eq, eq1, eq2, eq3, eq4, koef];
Subscript[P, k_][t_] = Subscript[f, k] + (t - Subscript[x, k])*(Subscript[A, k]*t + Subscript[B, k]);
eq1 = Table[Subscript[P, k][Subscript[x, k + 1]] == Subscript[P, k + 1][Subscript[x, k + 1]], {k, 0, n - 1}];
Subscript[PP, k_][t_] = D[Subscript[P, k][t], t];
eq2 = Table[Subscript[PP, k][Subscript[x, k + 1]] == Subscript[PP, k + 1][Subscript[x, k + 1]], {k, 0, n - 1}];
eq3 = {Subscript[P, n - 1][Subscript[x, n]] == Subscript[f, n]};
eq4 = {Subscript[PP, 0][Subscript[x, 0]] == 0};
eq = Join[eq1, eq2, eq3, eq4];
koef = Solve[eq, {}];
spl2 = Table[Subscript[P, k][t] /. koef, {k, 0, n - 1}] // Expand // Flatten
Table[(spl2[[k]] /. t -> Subscript[x, k - 1]) == Subscript[f, k - 1], {k, 1, n}]
(spl2[[n]] /. t -> Subscript[x, n]) == Subscript[f, n]
(D[spl2[[1]], {t, 1}] /. t -> Subscript[x, 0]) == 0
{4 + 2 (3 A0 + B0) == -20, -20 + 2 (5 A1 + B1) == 45,
 5 A0 + B0 == 3 A1 + B1, 7 A1 + B1 == 5 A2 + B2, 45 + 3 (8 A2 + B2) == 76, A0 + B0 == 0}

{{A0 -> -6, A1 ->  $\frac{113}{4}$ , A2 ->  $-\frac{236}{9}$ , B0 -> 6, B1 ->  $-\frac{435}{4}$ , B2 ->  $\frac{1981}{9}$ }}

{-2 + 12 t - 6 t2,  $\frac{1225}{4} - \frac{387 t}{2} + \frac{113 t^2}{4}$ ,  $-\frac{9500}{9} + \frac{3161 t}{9} - \frac{236 t^2}{9}$ }

{True, True, True}

True

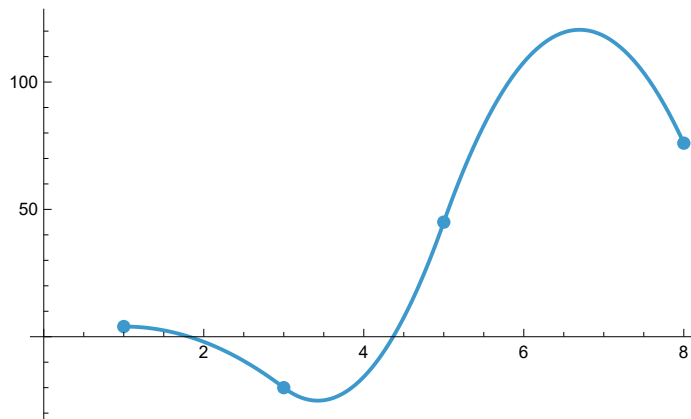
True

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GrC = Table[Plot[spl2[[k]], {t, Subscript[x, k - 1], Subscript[x, k]}], {k, 1, n}];
Show[GrA, GrC, PlotRange -> All]

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Кубический сплайн, $m = 3$, $S'(a) = 0$, $S''(b) = 0$

```

Clear[eq, eq1, eq2, eq3, eq4, eq5, eq6, koef];
a = Subscript[x, 0]; b = Subscript[x, n];
Subscript[P, k_][t_] = Subscript[f, k] + (t - Subscript[x, k])*(Subscript[A, k]*t^2 + Sub:
eq1 = Table[Subscript[P, k][Subscript[x, k + 1]] == Subscript[P, k + 1][Subscript[x, k + 1
Subscript[PP, k_][t_] = D[Subscript[P, k][t], t];
eq2 = Table[Subscript[PP, k][Subscript[x, k + 1]] == Subscript[PP, k + 1][Subscript[x, k +
Subscript[PPP, k_][t_] = D[Subscript[PP, k][t], t];
eq3 = Table[Subscript[PPP, k][Subscript[x, k + 1]] == Subscript[PPP, k + 1][Subscript[x, k
eq4 = {Subscript[P, n - 1][Subscript[x, n]] == Subscript[f, n]};
eq5 = {Subscript[PP, 0][Subscript[x, 0]] == 0};
eq6 = {Subscript[PPP, n - 1][Subscript[x, n]] == 0};
eq = Join[eq1, eq2, eq3, eq4, eq5, eq6]
koef = Solve[eq, {}]
spl3 = Table[Subscript[P, k][t] /. koef, {k, 0, n - 1}] // Expand // Flatten
Table[(spl3[[k]] /. t -> Subscript[x, k - 1]) == Subscript[f, k - 1], {k, 1, n}]
(spl3[[n]] /. t -> Subscript[x, n]) == Subscript[f, n]
(D[spl3[[1]], {t, 1}] /. t -> a) == 0
(D[spl3[[n]], {t, 2}] /. t -> b) == 0
{4 + 2 (9 A0 + 3 B0 + C0) == -20, -20 + 2 (25 A1 + 5 B1 + C1) == 45,
9 A0 + 3 B0 + 2 (6 A0 + B0) + C0 == 9 A1 + 3 B1 + C1,
25 A1 + 5 B1 + 2 (10 A1 + B1) + C1 == 25 A2 + 5 B2 + C2, 16 A0 + 2 B0 == 12 A1 + 2 B1,
24 A1 + 2 B1 == 20 A2 + 2 B2, 45 + 3 (64 A2 + 8 B2 + C2) == 76, A0 + B0 + C0 == 0, 38 A2 + 2 B2 == 0}

{ {A0 →  $\frac{511}{66}$ , A1 →  $-\frac{537}{88}$ , A2 →  $\frac{1537}{1188}$ , B0 →  $-\frac{1220}{33}$ ,
B1 →  $\frac{739}{12}$ , B2 →  $-\frac{29203}{1188}$ , C0 →  $\frac{643}{22}$ , C1 →  $-\frac{32435}{264}$ , C2 →  $\frac{3353}{27}$  } }

{  $-\frac{555}{22} + \frac{4369t}{66} - \frac{2951t^2}{66} + \frac{511t^3}{66}$ ,
 $\frac{30675}{88} - \frac{81209t}{264} + \frac{21091t^2}{264} - \frac{537t^3}{88}$ ,  $-\frac{15550}{27} + \frac{97849t}{396} - \frac{3074t^2}{99} + \frac{1537t^3}{1188}$  }

{True, True, True}

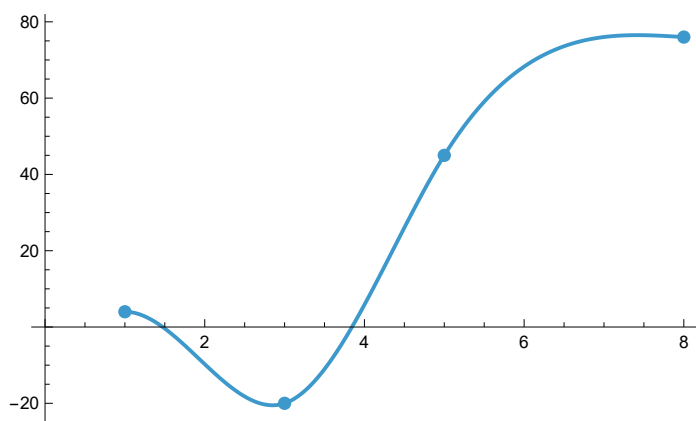
True

True

True

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GrD = Table[Plot[spl3[[k]], {t, Subscript[x, k - 1], Subscript[x, k]}], {k, 1, n}];
Show[GrA, GrD, PlotRange -> All]
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Все сплайны в одной системе координат;
 Show[GrA, GrB, GrC, GrD, PlotRange -> All]

